

## KCS

KCS stands for Knowledge-Centered Service, which is a methodology for creating and maintaining documentation as part of the service delivery process. KCS aims to improve the quality and efficiency of service organizations by capturing, structuring, and reusing the knowledge of service agents and customers. Some of the benefits of KCS are:

- Reduced resolution time and costs
- Increased customer satisfaction and loyalty
- Enhanced agent productivity and morale
- Improved service quality and consistency
- Increased organizational learning and innovation

KCS is based on the following principles:

- Integrate the creation and maintenance of knowledge into the service process
- Evolve the knowledge base based on demand and usage
- Develop a knowledge base of collective experience, not individual opinions
- Reward learning, collaboration, sharing, and improvement
- Value trust over control
- Enable people to reuse, improve, and flag knowledge
- Optimize the knowledge base for use and reuse
- Base decisions on data and facts

KCS has four main practices:

- Capture: Record the customer's context and the agent's experience in the knowledge base during the service interaction
- Structure: Use a standard format and metadata to make the knowledge articles easy to find and use
- Reuse: Search the knowledge base before solving a new issue and use existing knowledge articles to resolve or escalate the issue
- Improve: Update, flag, or archive knowledge articles based on feedback and usage patterns

KCS also has a maturity model that describes the stages of adoption and improvement of the methodology, from initial to optimized. The maturity model helps service organizations assess their current state, identify gaps and opportunities, and plan for the next steps. The maturity model consists of the following levels:

- Level 1: React - The service organization is reactive and focused on solving individual issues. There is no formal knowledge management process or system.
- Level 2: Capture - The service organization starts to capture and structure knowledge during the service process. There is a basic knowledge management system and some incentives for knowledge creation and reuse.

- Level 3: Structure - The service organization improves the quality and consistency of the knowledge base. There are clear roles and responsibilities for knowledge management and a feedback loop for knowledge improvement.
- Level 4: Reuse - The service organization leverages the knowledge base to resolve and prevent issues. There are metrics and rewards for knowledge reuse and improvement. The knowledge base is integrated with other systems and channels.
- Level 5: Optimize - The service organization uses the knowledge base to drive continuous improvement and innovation. There are advanced analytics and automation for knowledge management. The knowledge base is aligned with the organizational strategy and goals.

## **Unit 1 - Brief history of microbiology, Types of microorganisms, Basic idea of domain bacteria, proteobacteria, non proteobacteria Gram –ve and Gram +ve bacteria, lichens, algae, protozoa, helminthes, viral structures, viral multiplication, Role of microorganisms in the production of industrial chemicals and pharmaceuticals**

- Microbiology is the study of microscopic organisms, such as bacteria, viruses, fungi, protozoa, and algae.
- Microbiology essentially began with the development of the microscope in the 17th century.
- Some of the pioneers of microbiology are:
  - Antonie van Leeuwenhoek (1632-1723), a Dutch draper who was the first to observe and document various microorganisms, such as bacteria, protozoa, and fungi, using a microscope of his own design .
  - Robert Hooke (1635-1703), an English naturalist who coined the term “cell” and observed fungal structures and plant cells under the microscope .
  - Ferdinand Cohn (1828-1898), a German botanist who founded the field of bacteriology and classified bacteria into four groups based on their shape and motility .
  - Louis Pasteur (1822-1895), a French chemist who demonstrated the role of microbes in fermentation, disproved the theory of spontaneous generation, and developed the process of pasteurization and the first vaccines against anthrax and rabies.
  - Robert Koch (1843-1910), a German physician who established the germ theory of disease, identified the causative agents of anthrax, tuberculosis, and cholera, and developed methods for isolating and

- culturing bacteria.
- Microorganisms can be classified into different types based on their structure, function, and evolutionary relationships. Some of the major types of microorganisms are:
    - Bacteria, which are prokaryotic cells that lack a nucleus and membrane-bound organelles. They can be found in almost every environment and play important roles in nutrient cycling, decomposition, symbiosis, and pathogenesis. Bacteria are divided into two domains: Archaea and Bacteria.
    - Proteobacteria, which are a large and diverse group of gram-negative bacteria that belong to the domain Bacteria. They include many medically and environmentally important genera, such as *Escherichia*, *Salmonella*, *Vibrio*, *Helicobacter*, and *Rhizobium*.
    - Non-proteobacteria gram-negative bacteria, which are a diverse group of bacteria that belong to the domain Bacteria but are not classified as proteobacteria. They include many photosynthetic, nitrogen-fixing, and extremophilic bacteria, such as Cyanobacteria, Spirochaetes, Chlamydiae, and Bacteroidetes.
    - Gram-positive bacteria, which are a group of bacteria that belong to the domain Bacteria and have a thick cell wall that retains the purple stain in the Gram staining technique. They include many beneficial and pathogenic bacteria, such as *Bacillus*, *Clostridium*, *Staphylococcus*, *Streptococcus*, and *Lactobacillus*.
    - Lichens, which are symbiotic associations of fungi and algae or cyanobacteria. They can grow on various substrates, such as rocks, trees, and soil, and are important indicators of environmental quality and sources of natural products.
    - Algae, which are eukaryotic organisms that can perform photosynthesis and produce oxygen. They can be unicellular or multicellular and can be found in freshwater, marine, and terrestrial habitats. They include many groups, such as green algae, red algae, brown algae, and diatoms.
    - Protozoa, which are eukaryotic organisms that can move and feed on other organisms or organic matter. They can be unicellular or multicellular and can be found in aquatic and soil environments. They include many groups, such as amoebae, flagellates, ciliates, and apicomplexans.
    - Helminthes, which are eukaryotic organisms that have a worm-like body and can be parasitic or free-living. They can be multicellular and have complex organ systems. They include many groups, such as nematodes, trematodes, cestodes, and acantho

## Brief history of microbiology

- Microbiology is the scientific study of microorganisms, which are usually too small to be seen with the naked eye.
- Microbiology has its origins in the 16th century, when some people observed the existence of invisible living things that caused diseases or fermentation.
- The development of the microscope in the 17th century enabled the direct observation and documentation of microorganisms by scientists such as Robert Hooke and Anton van Leeuwenhoek.
- Robert Hooke (1635-1703) was an English scientist who coined the term “cell” after observing cork and plant tissues under a microscope. He also observed strands of fungi and microfossils.
- Anton van Leeuwenhoek (1632-1723) was a Dutch draper and amateur microscopist who made his own lenses and microscopes. He was the first to observe and describe bacteria, protozoa, algae, and other microorganisms, which he called “animalcules”.
- In the 18th and 19th centuries, microbiology became a more established and diverse field, with contributions from many scientists such as Louis Pasteur, Robert Koch, Ferdinand Cohn, Martinus Beijerinck, and Sergei Winogradsky.
- Louis Pasteur (1822-1895) was a French chemist and microbiologist who demonstrated the role of microorganisms in fermentation, spoilage, and disease. He also developed the germ theory of disease, the process of pasteurization, and the first vaccines against anthrax and rabies.
- Robert Koch (1843-1910) was a German physician and microbiologist who established the criteria for proving the causal relationship between a microbe and a disease, known as Koch’s postulates. He also identified the causative agents of anthrax, tuberculosis, and cholera, and developed methods for isolating and culturing bacteria.
- Ferdinand Cohn (1828-1898) was a German botanist and microbiologist who founded the field of bacteriology. He classified bacteria into four groups based on their shape and mode of reproduction, and discovered bacterial spores and endospores. He also studied algae and cyanobacteria, and described the life cycle of *Bacillus subtilis*.
- Martinus Beijerinck (1851-1931) was a Dutch microbiologist and botanist who pioneered the use of enrichment cultures and selective media to isolate and study specific types of microorganisms from natural environments. He also coined the term “virus” and discovered the first virus, the tobacco mosaic virus.
- Sergei Winogradsky (1856-1953) was a Russian microbiologist and ecologist who studied the role of microorganisms in biogeochemical cycles, such as nitrogen fixation, nitrification, and sulfur oxidation. He also developed the concept of chemolithotrophy, the use of inorganic compounds as energy sources by some bacteria.

## Types of microorganisms

Microorganisms are microscopic organisms that exist as unicellular, multicellular, or cell clusters. They can be beneficial or harmful to life, and they can be divided into six major types :

- **Bacteria:** These are prokaryotic cells, meaning they lack a nucleus and other membrane-bound organelles. They have a cell wall made of peptidoglycan, and they can be classified by their shape, Gram stain, oxygen requirement, and metabolic diversity. Some examples of bacteria are *Escherichia coli*, *Staphylococcus aureus*, and *Streptococcus pyogenes*.
- **Archaea:** These are also prokaryotic cells, but they differ from bacteria in their cell wall composition, membrane lipids, and ribosomal RNA sequences. They are often found in extreme environments, such as hot springs, salt lakes, and deep-sea vents. Some examples of archaea are *Methanobacterium*, *Halobacterium*, and *Thermococcus*.
- **Protozoa:** These are eukaryotic cells, meaning they have a nucleus and other membrane-bound organelles. They are usually motile and heterotrophic, meaning they obtain their nutrients by ingesting other organisms or organic matter. They can be classified by their mode of locomotion, such as flagella, cilia, or pseudopodia. Some examples of protozoa are *Paramecium*, *Amoeba*, and *Plasmodium*.
- **Algae:** These are also eukaryotic cells, but they are photosynthetic, meaning they use light energy to synthesize organic molecules. They have a cell wall made of cellulose, and they can be unicellular or multicellular. They can be classified by their pigments, such as chlorophyll, carotenoids, or phycobilins. Some examples of algae are *Chlamydomonas*, *Spirogyra*, and *Ulva*.
- **Fungi:** These are also eukaryotic cells, but they are saprophytic, meaning they decompose organic matter. They have a cell wall made of chitin, and they can be unicellular or multicellular. They can be classified by their morphology, such as yeasts, molds, or mushrooms. Some examples of fungi are *Saccharomyces*, *Aspergillus*, and *Agaricus*.
- **Viruses:** These are not cells, but acellular particles that consist of a nucleic acid (DNA or RNA) surrounded by a protein coat. They are obligate intracellular parasites, meaning they can only replicate inside a host cell. They can be classified by their shape, size, genome type, and host range. Some examples of viruses are Influenza, Herpes, and HIV.

## Basic idea of domain bacteria

- Domain bacteria is one of the three domains of life, along with domain archaea and domain eukarya.
- Domain bacteria includes the prokaryotes people encounter on an everyday basis. Prokaryotes are organisms that lack a membrane-bound nucleus and other organelles.

- Most bacterial species are heterotrophic; that is, they acquire their food from organic matter. The largest number of bacteria are saprobic, meaning that they feed on dead or decaying organic matter. A few bacterial species are parasitic; these bacteria live within host organisms and cause disease.
- Domain bacteria can be classified into five main phyla: proteobacteria, cyanobacteria, firmicutes, actinobacteria, and spirochaetes.
- Proteobacteria are the largest and most diverse group of bacteria. They include many common pathogens, such as *E. coli*, *Salmonella*, and *H. pylori*, as well as many beneficial bacteria, such as nitrogen-fixing bacteria and bioluminescent bacteria.
- Cyanobacteria are also known as blue-green algae. They are capable of photosynthesis and produce oxygen. They are important for the evolution of life on Earth, as they are believed to be the first organisms to perform oxygenic photosynthesis.
- Firmicutes are mostly gram-positive bacteria, meaning that they have a thick cell wall that retains a purple dye in a staining technique. They include many bacteria that live in the human gut, such as *Lactobacillus* and *Clostridium*, as well as some pathogens, such as *Staphylococcus* and *Streptococcus*.
- Actinobacteria are also mostly gram-positive bacteria, but they have a high GC content in their DNA, meaning that they have more guanine and cytosine bases than adenine and thymine bases. They include many soil bacteria, such as *Streptomyces*, which produce antibiotics, and *Mycobacterium*, which cause tuberculosis and leprosy.
- Spirochaetes are bacteria that have a spiral shape and a flagellum that allows them to move. They include some pathogens, such as *Treponema pallidum*, which causes syphilis, and *Borrelia burgdorferi*, which causes Lyme disease.
- Domain bacteria play a vital role in the production of industrial chemicals and pharmaceuticals, such as ethanol, acetone, antibiotics, insulin, and vaccines. They also participate in various biogeochemical cycles, such as the carbon cycle, the nitrogen cycle, and the sulfur cycle, and help maintain the balance of the environment .

## Proteobacteria

- Proteobacteria is a phylum of gram-negative bacteria discovered by Carl Woese in the 1980s based on nucleotide sequence homology .
- Proteobacteria are further classified into the classes alpha-, beta-, gamma-, delta- and epsilonproteobacteria, each class having separate orders, families, genera, and species .
- Proteobacteria include a wide variety of pathogenic and free-living bacteria, such as *Escherichia*, *Salmonella*, *Vibrio*, *Yersinia*, *Legionella*, and many others .

- Proteobacteria also include many of the bacteria responsible for nitrogen fixation, a process that converts atmospheric nitrogen into ammonia, which can be used by plants and other organisms .
- Proteobacteria are diverse in their metabolic and ecological adaptations, such as aerobic or anaerobic respiration, fermentation, photosynthesis, chemolithotrophy, and symbiosis .
- Some examples of proteobacteria are:
  - Alphaproteobacteria: oligotrophs, organisms capable of living in low-nutrient environments such as deep oceanic sediments, glacial ice, or deep undersurface soil. Some examples are *Rhizobium*, *Agrobacterium*, *Rickettsia*, and *Caulobacter*.
  - Betaproteobacteria: mostly aerobic or facultatively anaerobic bacteria that can use various organic and inorganic compounds as electron donors or acceptors. Some examples are *Neisseria*, *Bordetella*, *Nitrosomonas*, and *Burkholderia*.
  - Gammaproteobacteria: the largest and most diverse class of proteobacteria, comprising many medically and environmentally important bacteria. Some examples are *Pseudomonas*, *Escherichia*, *Salmonella*, *Vibrio*, and *Legionella*.
  - Deltaproteobacteria: mostly anaerobic bacteria that can perform sulfate reduction, a process that converts sulfate into hydrogen sulfide, which can be toxic or beneficial depending on the context. Some examples are *Desulfovibrio*, *Bdellovibrio*, and *Myxococcus*.
  - Epsilonproteobacteria: mostly microaerophilic or anaerobic bacteria that can inhabit extreme environments such as hot springs, hydrothermal vents, or animal guts. Some examples are *Helicobacter*, *Campylobacter*, and *Wolinella*.

## Nonproteobacteria Gram-Negative Bacteria

- Nonproteobacteria are a diverse group of bacteria that do not belong to the phylum Proteobacteria, which contains the majority of gram-negative bacteria.
- Nonproteobacteria can be either gram-negative or gram-positive, depending on the presence or absence of an outer membrane in their cell wall.
- Nonproteobacteria can have various metabolic modes, such as photoautotrophy, chemoheterotrophy, or chemoautotrophy.
- Nonproteobacteria can be either motile or nonmotile, and some have distinctive shapes, such as spirals, rods, or filaments.
- Some examples of nonproteobacteria are:

- Chlamydia: a group of obligate intracellular parasites that cause various diseases in humans and animals, such as chlamydia, trachoma, and psittacosis. They have a unique developmental cycle that involves two forms: elementary bodies and reticulate bodies.
- Spirochetes: a group of motile, spiral-shaped bacteria that have a flexible cell wall and a unique flagellar arrangement called axial filaments. They can cause diseases such as syphilis, Lyme disease, and leptospirosis. They are difficult or impossible to culture in the laboratory.
- CFB group: a group of gram-negative bacteria that includes the genera *Cytophaga*, *Fusobacterium*, and *Bacteroides*. They share some similarities in their DNA sequences, but are phylogenetically diverse. They are mostly found in the oral cavity, the gastrointestinal tract, and the soil. They can be involved in periodontal disease, abscesses, and biodegradation of organic matter.
- Planctomycetes: a group of gram-negative bacteria that have unusual features, such as a lack of peptidoglycan in their cell wall, a compartmentalized cytoplasm, and budding reproduction. They are mostly aquatic and can perform anaerobic ammonium oxidation (anammox), a process that contributes to the nitrogen cycle.
- Phototrophic bacteria: a group of bacteria that can use light as an energy source for photosynthesis. They can be either gram-negative or gram-positive, and can have different types of pigments, such as chlorophylls, bacteriochlorophylls, or carotenoids. They can be divided into two main groups: oxygenic and anoxygenic. Oxygenic phototrophic bacteria produce oxygen as a byproduct of photosynthesis, while anoxygenic phototrophic bacteria do not. Some examples of phototrophic bacteria are cyanobacteria, purple bacteria, green bacteria, and heliobacteria.

## Gram-positive bacteria

- Gram-positive bacteria are bacteria that have a thick cell wall composed of peptidoglycan and teichoic acids .
- Gram-positive bacteria can be distinguished from gram-negative bacteria by the Gram stain test, which involves a chemical dye that stains the cell wall purple or blue .
- Gram-positive bacteria include many important pathogens, such as *Staphylococcus*, *Streptococcus*, *Bacillus*, *Clostridium*, *Corynebacterium*, *Listeria*, and *Mycobacterium* .
- Gram-positive bacteria can cause various infections, such as skin infections, pneumonia, meningitis, anthrax, tetanus, diphtheria, tuberculosis, and leprosy .
- Gram-positive bacteria can also produce useful substances, such as antibiotics, enzymes, vitamins, and probiotics.

- Gram-positive bacteria can undergo bacterial transformation, which is the uptake and incorporation of DNA from the environment.
- Gram-positive bacteria can be treated with antibiotics that target the cell wall, such as penicillins, cephalosporins, vancomycin, and daptomycin .
- Gram-positive bacteria are classified into two major groups based on their cell shape and arrangement: cocci (spherical) and bacilli (rod-shaped) .
- Gram-positive cocci can be further divided into staphylococci (clusters), streptococci (chains), and enterococci (pairs or short chains) .
- Gram-positive bacilli can be further divided into spore-forming (such as Bacillus and Clostridium) and non-spore-forming (such as Corynebacterium and Listeria) .

## Lichens

- Lichens are **plantlike organisms** that consist of a **symbiotic association** of algae (usually green) or cyanobacteria and fungi (mostly ascomycetes and basidiomycetes).
- Lichens are **composite organisms** that arise from algae or cyanobacteria living among filaments of multiple fungi species in a **mutualistic relationship**.
- The dominant partner is the **fungus**, which gives the lichen the majority of its characteristics, from its **thallus shape** to its **fruiting bodies**.
- The alga can be either a **green alga** or a **blue-green alga**, otherwise known as **cyanobacteria**.
- Lichens are found **worldwide** and occur in a variety of **environmental conditions**.
- Lichens are **pioneer species**, among the first living things to grow on **bare rock** or areas denuded of life by a **disaster**.
- Lichens may have to compete with plants for access to **sunlight**, but because of their small size and slow growth, they thrive in places where higher plants have difficulty growing.
- Lichens have various **roles** in the production of **industrial chemicals** and **pharmaceuticals**, such as dyes, perfumes, antibiotics, antivirals, anti-inflammatories, and anticancer agents.

## Algae

- Algae are a large and diverse group of photosynthetic eukaryotic organisms that live in aquatic or moist environments .
- Algae are not plants, but they share some characteristics with them, such as having chlorophyll, cell walls, and the ability to produce oxygen .
- Algae can be classified into different groups based on their pigments, cell structure, life cycle, and evolutionary history .
- Some of the major groups of algae are:

- Green algae: They have chlorophyll a and b, and store starch in their chloroplasts. They are the closest relatives of plants and can be unicellular or multicellular .
- Brown algae: They have chlorophyll a and c, and a brown pigment called fucoxanthin. They are mostly multicellular and form large seaweeds, such as kelp .
- Red algae: They have chlorophyll a and d, and a red pigment called phycoerythrin. They are mostly multicellular and live in deep or warm waters .
- Diatoms: They have chlorophyll a and c, and a golden-brown pigment called diatoxanthin. They are unicellular and have silica cell walls that form intricate patterns .
- Dinoflagellates: They have chlorophyll a and c, and various other pigments. They are unicellular and have two flagella that make them spin. Some of them can produce bioluminescence or toxins .
- Euglenoids: They have chlorophyll a and b, and a red eyespot that helps them sense light. They are unicellular and have a flexible pellicle that allows them to change shape. Some of them can lose their chloroplasts and become heterotrophic .
- Cyanobacteria: They are prokaryotic organisms that have chlorophyll a and phycobilins. They are also known as blue-green algae, although they are not true algae. They can form colonies or filaments, and some of them can fix nitrogen .
- Algae play an important role in the production of industrial chemicals and pharmaceuticals, such as agar, carrageenan, alginates, iodine, biofuels, antioxidants, vitamins, and antibiotics .

## Protozoa

- Protozoa are a group of single-celled eukaryotic organisms that can be free-living or parasitic, and that feed on organic matter such as other microorganisms or organic tissues and debris .
- Protozoa were formerly considered a phylum or sub-kingdom of the animal kingdom, but are now classified into various phyla or supergroups based on their morphology, motility, and phylogeny.
- Some of the characteristics of protozoa are:
  - They are microscopic and acellular, meaning they do not have tissues or organs.
  - They have one or more nuclei, which may be simple or complex, and may contain multiple chromosomes.
  - They have various modes of locomotion, such as flagella, cilia, pseudopodia, or gliding.
  - They have diverse modes of nutrition, such as autotrophy, heterotrophy, mixotrophy, or symbiosis.
  - They have complex life cycles, which may involve asexual and sexual

- reproduction, and may include different stages such as trophozoites, cysts, spores, or gametes.
  - They have various adaptations to survive in different environments, such as osmoregulation, encystment, bioluminescence, or biomineralization.
- Some of the major phyla or supergroups of protozoa are:
  - Excavata, which includes flagellates such as Giardia, Trypanosoma, and Euglena.
  - Amoebozoa, which includes amoebas and slime molds.
  - Rhizaria, which includes radiolarians, foraminiferans, and cercozoans.
  - Alveolata, which includes ciliates, dinoflagellates, and apicomplexans.
  - Stramenopila, which includes diatoms, brown algae, and water molds.
  - Opisthokonta, which includes choanoflagellates and fungi.
  - Archaeplastida, which includes red algae and green algae.
- Protozoa play important roles in the production of industrial chemicals and pharmaceuticals, such as:
  - Ethanol, which is produced by some flagellates and ciliates from sugar or starch.
  - Antibiotics, such as penicillin, which is derived from a fungus.
  - Antimalarials, such as quinine, which is extracted from a plant that contains a protozoan endosymbiont.
  - Antiprotozoals, such as metronidazole, which is effective against some anaerobic protozoa.

## Helminthes

- Helminthes are **worm-like parasites** that can infect humans and other animals.
- They are **invertebrates** characterized by elongated, flat or round bodies.
- They are classified into three major groups based on their shape and habitat :
  - **Flukes (trematodes)**: flat, leaf-shaped worms that live in the liver, lungs, blood vessels, or intestines of the host .
  - **Tapeworms (cestodes)**: flat, ribbon-like worms that live in the intestines of the host and have a scolex (head) with hooks and suckers for attachment .
  - **Roundworms (nematodes)**: cylindrical, thread-like worms that live in the intestines, blood, lymphatic system, or tissues of the host and have a complete digestive system .
- Helminthes have different modes of reproduction depending on their type and species:
  - Some are **hermaphroditic**, meaning they have both male and female reproductive organs and can self-fertilize or cross-fertilize with other

worms of the same species.

- Some are **bisexual**, meaning they have separate male and female individuals and require mating for fertilization.
- Helminthes have different stages of growth and development depending on their type and species:
  - Some have a **direct life cycle**, meaning they only require one host to complete their development and transmission.
  - Some have an **indirect life cycle**, meaning they require two or more hosts (intermediate and definitive) to complete their development and transmission.
- Helminthes can cause various diseases and symptoms in humans and animals, such as intestinal obstruction, malnutrition, anemia, allergic reactions, tissue damage, inflammation, and organ failure .
- Helminthes can be diagnosed by microscopic examination of stool, blood, urine, or tissue samples, or by serological tests for antibodies or antigens .
- Helminthes can be treated by anthelmintic drugs that target their metabolism, nervous system, or reproductive system, or by surgical removal in some cases .
- Helminthes can be prevented by improving sanitation, hygiene, and vector control, or by vaccination in some cases .

## Viral Structures

- Viruses are noncellular genetic elements that use a living cell for their replication and have an extracellular state.
- Viruses are ultramicroscopic particles containing nucleic acid surrounded by protein, and in some cases, other macromolecular components such as a membranelike envelope.
- Viruses are observed in a wide range of shapes and sizes, but these shapes are constant in a particular viral family.
- The basic structure of all viruses includes nucleic acid, either DNA (Deoxyribonucleic Acid) or RNA (Ribonucleic Acid) called a genome which is enveloped by a protein coat called a capsid .
- Many viruses have a lipoprotein bilayer that encloses the capsid, called an envelope .
- The nucleic acid of a virus contains the genetic information for the synthesis of proteins and replication.
- The capsid is composed of subunits called capsomeres, which may have different shapes and arrangements depending on the type of virus .
- The envelope may have spikes or glycoproteins that help the virus attach to the host cell or evade the host immune system .
- Viruses can be classified into two main classes based on their nucleic acid: RNA viruses and DNA viruses.
- RNA viruses have single-stranded or double-stranded RNA as their genome, and they may be positive-sense or negative-sense, depending on

the polarity of their RNA .

- DNA viruses have single-stranded or double-stranded DNA as their genome, and they may be linear or circular, depending on the shape of their DNA .
- Some examples of RNA viruses are influenza virus, coronavirus, poliovirus, and HIV .
- Some examples of DNA viruses are herpesvirus, adenovirus, poxvirus, and hepatitis B virus .
- Viruses can also be classified into different types based on their shape, such as helical, icosahedral, complex, or enveloped .
- Helical viruses have a rod-shaped or filamentous capsid that forms a helix around the nucleic acid .
- Icosahedral viruses have a spherical capsid that consists of 20 triangular faces and 12 vertices .
- Complex viruses have a capsid that is neither helical nor icosahedral, and may have additional structures such as a tail or a head .
- Enveloped viruses have a capsid that is surrounded by a lipid bilayer derived from the host cell membrane .
- Some examples of helical viruses are tobacco mosaic virus, rabies virus, and Ebola virus .
- Some examples of icosahedral viruses are poliovirus, adenovirus, and herpesvirus .
- Some examples of complex viruses are bacteriophage, poxvirus, and mimivirus .
- Some examples of enveloped viruses are influenza virus, coronavirus, and HIV .

## Viral Multiplication

Viral multiplication is the process of producing new viruses using the host cell's machinery and resources. Viruses are obligate intracellular parasites, which means they can only replicate inside living cells. Viruses have different types of genomes, such as DNA or RNA, single-stranded or double-stranded, linear or circular, etc. The type of genome determines the mode of replication and the enzymes required for viral multiplication.

The general steps of viral multiplication are:

- **Attachment:** The virus binds to a specific receptor on the surface of the host cell. The receptor is usually a protein or a carbohydrate that is involved in normal cellular functions. The attachment is mediated by viral proteins called attachment proteins or spikes, which recognize and bind to the receptor. The attachment determines the host range and tissue tropism of the virus, i.e., the types of cells and organisms that the virus can infect.
- **Penetration:** The virus enters the host cell by one of the following mech-

anisms: endocytosis, fusion, or injection. Endocytosis is the process of engulfing the virus by the cell membrane and forming a vesicle called an endosome. Fusion is the process of merging the viral envelope with the cell membrane and releasing the viral capsid into the cytoplasm. Injection is the process of injecting the viral genome into the cell through a tail-like structure, as in some bacteriophages.

- **Uncoating:** The virus releases its genome from the capsid and makes it available for replication and transcription. Uncoating can occur in the endosome, in the cytoplasm, or in the nucleus, depending on the type of virus. Uncoating may require the action of viral or cellular enzymes, or changes in pH or temperature.
- **Biosynthesis:** The virus synthesizes new copies of its genome and viral proteins using the host cell's machinery and resources. The biosynthesis can be divided into two phases: early and late. The early phase involves the expression of viral genes that encode enzymes and regulatory proteins that are essential for viral replication and transcription. The late phase involves the expression of viral genes that encode structural proteins that form the viral capsid and envelope. The biosynthesis depends on the type of viral genome and the location of replication and transcription. For example, DNA viruses usually replicate and transcribe in the nucleus, while RNA viruses usually replicate and transcribe in the cytoplasm. Some viruses use reverse transcriptase to convert their RNA genome into DNA, such as retroviruses and hepadnaviruses.
- **Assembly:** The virus assembles new virions by putting together the viral genome and the viral proteins. The assembly can occur in the cytoplasm, in the nucleus, or at the cell membrane, depending on the type of virus. The assembly may require the help of viral or cellular proteins, or self-assembly of viral components.
- **Release:** The virus exits the host cell by one of the following mechanisms: lysis, budding, or exocytosis. Lysis is the process of breaking the cell membrane and releasing the virions into the extracellular environment. Lysis usually results in the death of the host cell. Budding is the process of pinching off a portion of the cell membrane and wrapping it around the virion, forming a viral envelope. Budding may or may not result in the death of the host cell. Exocytosis is the process of fusing the vesicle containing the virion with the cell membrane and releasing it into the extracellular environment. Exocytosis usually does not result in the death of the host cell.

The viral multiplication can be classified into two types based on the life cycle, i.e., lytic cycle or lysogenic cycle. The lytic cycle is the process of producing new virions and destroying the host cell. The lysogenic cycle is the process of integrating the viral genome into the host genome and remaining dormant until a trigger activates the lytic cycle. Some viruses can undergo both cycles, such as bacteriophages and herpesviruses. Some viruses can only undergo the lytic cycle, such as polioviruses and influenza viruses. Some viruses can only undergo

the lysogenic cycle, such as retroviruses and proviruses.

The viral multiplication can be measured by the one-step multiplication curve, which shows the number of infectious virions over time after infecting a host cell. The one-step multiplication curve follows three steps: inoculation, eclipse, and burst. Inoculation is the period when the virions attach to the host cells. Eclipse is the period when the virions enter the host cells and uncoat their

## **Role of microorganisms in the production of industrial chemicals and pharmaceuticals**

- Microorganisms are used in industry to produce a variety of organic compounds including acids, growth stimulants, and enzymes.
- Microorganisms can be genetically modified or engineered to aid in large-scale production.
- Microorganisms can also use carbon dioxide and hydrogen as feedstocks to produce industrial chemicals and fuels.
- Some examples of industrial chemicals and pharmaceuticals produced by microorganisms are:
  - Amino acids: Microbes make only the L-isomer, which is the biologically active form of amino acids. Amino acids are used as food additives, nutritional supplements, and precursors for pharmaceuticals.
  - Citric acid: Produced by a mold called *Aspergillus niger*. Citric acid is used as a flavoring agent, preservative, and antioxidant in food and beverages.
  - Enzymes: Produced by various bacteria, fungi, and yeasts. Enzymes are used to catalyze biochemical reactions, such as starch hydrolysis, protein digestion, and DNA replication. Enzymes are used in food processing, detergent manufacturing, biotechnology, and medicine.
  - Antibiotics: Produced by bacteria and fungi that inhibit the growth of other microorganisms. Antibiotics are used to treat bacterial infections and prevent the spread of diseases.
  - Vaccines: Produced by attenuated or killed microorganisms that stimulate the immune system to produce antibodies against specific pathogens. Vaccines are used to prevent infectious diseases and protect public health.

## Unit 2 - Functional Anatomy of Prokaryotic and Eukaryotic Cells: Size, shape, and arrangement of bacterial cells.

### Structure and function of cells

- Prokaryotic cells are cells that lack a nucleus and other membrane-bound organelles. They are usually smaller and simpler than eukaryotic cells, which have a nucleus and other membrane-bound organelles.
- Eukaryotic cells are cells that have a nucleus and other membrane-bound organelles. They are usually larger and more complex than prokaryotic cells, which lack a nucleus and other membrane-bound organelles.
- Bacterial cells are prokaryotic cells that belong to the domain Bacteria. They have a variety of shapes, sizes, and arrangements, depending on their species and environmental conditions.
- The most common shapes of bacterial cells are:
  - Coccus: spherical or oval
  - Bacillus: rod-shaped or cylindrical
  - Spirillum: spiral or curved
  - Vibrio: comma-shaped
  - Coccobacillus: short and plump
  - Spirochete: flexible and helical
- The most common arrangements of bacterial cells are:
  - Single: individual cells
  - Diplo: pairs of cells
  - Strepto: chains of cells
  - Staphylo: clusters of cells
  - Tetrads: groups of four cells
  - Sarcinae: groups of eight cells
  - Palisades: side-by-side rows of cells
- The structure and function of bacterial cells are determined by their cell envelope, which consists of three layers: the cell wall, the cell membrane, and the capsule.
  - The cell wall is a rigid layer that surrounds the cell membrane and provides shape, support, and protection to the cell. It also prevents the cell from bursting due to osmotic pressure. The cell wall is composed of peptidoglycan, a polymer of sugar and amino acids, and may also contain other substances, such as lipopolysaccharide, teichoic acid, or mycolic acid, depending on the type of bacteria. Bacteria can be classified into two groups based on the structure and composition of their cell wall: gram-positive and gram-negative.
    - \* Gram-positive bacteria have a thick layer of peptidoglycan that retains the purple dye of the Gram stain. They also have teichoic acid, which is a polymer of alcohol and phosphate, attached to the peptidoglycan. Gram-positive bacteria are more susceptible to antibiotics that target the cell wall, such as penicillin and lysozyme.
    - \* Gram-negative bacteria have a thin layer of peptidoglycan that

does not retain the purple dye of the Gram stain. They also have an outer membrane, which is a bilayer of phospholipids and lipopolysaccharide, attached to the peptidoglycan by lipoproteins. The outer membrane acts as a barrier to many substances, including antibiotics, and contains porins, which are proteins that allow the passage of small molecules. The lipopolysaccharide, also known as endotoxin, is a complex molecule that consists of a lipid A, a core polysaccharide, and an O antigen. The lipid A is responsible for the toxic effects of gram-negative bacteria, such as fever and shock, while the O antigen is used for the identification of different strains of bacteria.

- The cell membrane is a semi-permeable layer that surrounds the cytoplasm and regulates the transport of materials in and out of the cell. It is composed of a phospholipid bilayer with embedded proteins, such as transporters, receptors, and enzymes. The cell membrane also contains hopanoids, which are sterol-like molecules that stabilize the membrane and make it more resistant to heat and pH changes. The cell membrane is involved in many cellular processes, such as respiration, photosynthesis, secretion, and signal transduction.
- The capsule is a sticky layer that surrounds the cell wall and the cell membrane and protects the cell from dehydration, phagocytosis, and attachment by viruses or other cells. It is composed of polysaccharides, proteins, or both, and may vary in thickness, composition, and structure, depending on the type of bacteria. The capsule also helps the bacteria adhere to surfaces and form biofilms, which are communities of bacteria that are embedded in a matrix of extracellular polymeric substances. Biofilms can enhance the survival and virulence of bacteria, as well as their resistance to antibiotics and host defenses.

## Functional Anatomy of Prokaryotic and Eukaryotic Cells

- Prokaryotic and eukaryotic cells are the two basic types of cells that make up all living organisms.
- Prokaryotic cells are simpler and smaller than eukaryotic cells, and lack a nucleus and other membrane-bound organelles.
- Eukaryotic cells are more complex and larger than prokaryotic cells, and have a nucleus and other membrane-bound organelles that perform specific functions.
- The main differences between prokaryotic and eukaryotic cells are summarized in the table below:

| Feature       | Prokaryotic cells                                                               | Eukaryotic cells                                                                                                 |
|---------------|---------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|
| Size          | 0.1-5 micrometers                                                               | 10-100 micrometers                                                                                               |
| Shape         | Mostly spherical, rod-shaped, or spiral                                         | Variable, depending on the type of cell                                                                          |
| Arrangement   | Single-celled or in chains, clusters, or filaments                              | Single-celled or multicellular                                                                                   |
| DNA           | Circular, in the cytoplasm, not associated with histones                        | Linear, in the nucleus, associated with histones                                                                 |
| Organelles    | No membrane-bound organelles, except ribosomes                                  | Many membrane-bound organelles, such as mitochondria, chloroplasts, endoplasmic reticulum, Golgi apparatus, etc. |
| Cell wall     | Present in most prokaryotes, composed of peptidoglycan or other polysaccharides | Present in some eukaryotes, such as plants and fungi, composed of cellulose or chitin                            |
| Cell membrane | Phospholipid bilayer with embedded proteins                                     | Phospholipid bilayer with embedded proteins and carbohydrates                                                    |
| Cell division | Binary fission, a simple process of splitting into two identical cells          | Mitosis or meiosis, complex processes of dividing the nucleus and the cytoplasm into two or four daughter cells  |
| Reproduction  | Asexual, by binary fission or budding                                           | Asexual or sexual, by mitosis or meiosis and fertilization                                                       |

- The structure and function of some of the common features of prokaryotic and eukaryotic cells are explained below:

## Plasma membrane

- The plasma membrane is the outermost layer of the cell that separates it from the external environment and regulates the movement of substances into and out of the cell.

- The plasma membrane is composed of a phospholipid bilayer, which consists of two layers of phospholipids, each with a hydrophilic (water-loving) head and a hydrophobic (water-fearing) tail.
- The phospholipid bilayer is fluid and flexible, and allows the membrane to change shape and adapt to different conditions.
- The plasma membrane also contains proteins and carbohydrates that are embedded in or attached to the phospholipid bilayer. These molecules perform various functions, such as transport, signaling, recognition, and adhesion.

## Cytoplasm

- The cytoplasm is the gel-like substance that fills the interior of the cell and contains the DNA, ribosomes, and other molecules and structures that are involved in the cell's metabolism, growth, and reproduction.
- The cytoplasm of prokaryotic cells is mostly water, with dissolved salts, sugars, amino acids, and other organic molecules. It also contains the nucleoid, which is the region where the circular DNA is located, and plasmids, which are small, circular, extra-chromosomal DNA molecules that can carry genes for antibiotic resistance, virulence, or other traits.
- The cytoplasm of eukaryotic cells is more complex and organized than that of prokaryotic cells. It contains the nucleus, which is the membrane-bound compartment that houses the linear DNA, and various organelles, which are specialized structures that perform specific functions for the cell. The cytoplasm also contains the cytoskeleton, which is a network of protein fibers that provide shape, support, and movement to the cell.

## Ribosomes

- Ribosomes are the structures that synthesize proteins from amino acids, using the information encoded in the DNA and the messenger RNA (mRNA).
- Ribosomes are composed of two subunits, one large and one small, each made of ribosomal RNA (rRNA) and proteins.
- Ribosomes can be found either free in the cytoplasm or attached to the endoplasmic reticulum (ER), a membrane-bound organelle that is involved in protein synthesis and modification.
- Ribosomes are similar in prokaryotic

## Size, Shape, and Arrangement of Bacterial Cells

- Bacterial cells are microscopic organisms that cannot be seen by the naked eye. They are measured in micrometers ( $\mu\text{m}$ ), which are one-thousandth of a millimeter. The average size of a bacterial cell is between 0.2 and 2.0  $\mu\text{m}$ .

- Bacterial cells have different shapes and arrangements that help them adapt to different environments and functions. The most common shapes are:
  - Cocci: spherical or oval cells that can be arranged in pairs, chains, clusters, or other patterns .
  - Bacilli: rod-shaped or cylindrical cells that can be arranged in singles, pairs, chains, or palisades .
  - Spirilla: spiral or curved cells that can be rigid or flexible, and can have one or more flagella for motility .
- Some bacterial cells have more complex shapes and arrangements, such as:
  - Vibrios: comma-shaped cells that are similar to spirilla but shorter and thicker .
  - Spirochetes: long, thin, and flexible cells that have a corkscrew-like motion and are often pathogenic .
  - Actinomycetes: filamentous or branched cells that resemble fungi and produce antibiotics .
  - Mycoplasmas: very small cells that lack a cell wall and have a pleomorphic shape that can change according to the environment .
  - Nanobacteria: extremely small cells that are controversial and may not be true bacteria. They have an average size of 323 nm by 242 nm and may scavenge nutrients from other bacteria.

## Size, shape, and arrangement of bacterial cells

- Bacterial cells are microscopic organisms that cannot be seen by the naked eye. They are measured in micrometers ( $\mu\text{m}$ ), which are one-thousandth of a millimeter. The average size of a bacterial cell is between 0.2 and 2.0  $\mu\text{m}$  .
- Bacterial cells have different shapes and arrangements that help them adapt to different environments and functions. The most common shapes are:
  - Cocci: spherical or oval-shaped bacteria. They can be arranged in pairs (diplococci), chains (streptococci), clusters (staphylococci), or tetrads (groups of four).
  - Bacilli: rod-shaped or cylindrical bacteria. They can be arranged in singles, pairs (diplobacilli), chains (streptobacilli), or palisades (parallel rows).
  - Spirilla: spiral-shaped or curved bacteria. They can be rigid (spirilla) or flexible (spirochetes). They are usually arranged in singles or loose coils.
- Some bacterial cells have special structures or appendages that modify their shape or arrangement, such as:
  - Capsules: slimy layers of polysaccharides or proteins that surround some bacterial cells and protect them from drying out, phagocytosis,

or antibiotics.

- Flagella: long, whip-like structures that protrude from some bacterial cells and enable them to move or swim.
- Pili: short, hair-like structures that extend from some bacterial cells and allow them to attach to surfaces, exchange genetic material, or scavenge nutrients .
- Some examples of bacterial cells with different size, shape, and arrangement are:
  - Staphylococcus aureus: a cocci bacterium that forms grape-like clusters and causes skin infections, food poisoning, and toxic shock syndrome.
  - Escherichia coli: a bacillus bacterium that is normally found in the human intestine and can cause diarrhea, urinary tract infections, or meningitis .
  - Treponema pallidum: a spirochete bacterium that causes syphilis, a sexually transmitted disease.
  - Mycoplasma genitalium: one of the smallest known bacteria, with a length of only 323 nm and a width of 242 nm. It has no cell wall and causes genital infections.

## Arrangement of Bacterial Cells

- Bacteria are microscopic, single-celled organisms that can have different shapes and arrangements.
- The shape of a bacterial cell is determined by its cell wall, which is made of peptidoglycan, a polymer of carbohydrates and proteins.
- The arrangement of bacterial cells is influenced by how they divide and how they stick together after division.
- The most common shapes of bacterial cells are:
  - Cocci: spherical or oval cells
  - Bacilli: rod-shaped or cylindrical cells
  - Spirilla: spiral or curved cells
- The most common arrangements of bacterial cells are:
  - Diplo: pairs of cells
  - Staphylo: grape-like clusters of cells
  - Strepto: chains of cells
  - Tetrad: groups of four cells
  - Sarcina: groups of eight cells
- Some examples of bacteria with different shapes and arrangements are:
  - Diplococci: Streptococcus pneumoniae, Neisseria gonorrhoeae
  - Staphylococci: Staphylococcus aureus, Staphylococcus epidermidis
  - Streptobacilli: Bacillus anthracis, Escherichia coli
  - Tetrads: Micrococcus luteus, Micrococcus roseus
  - Sarcinae: Sarcina ventriculi, Sarcina lutea
  - Vibrios: Vibrio cholerae, Vibrio parahaemolyticus

- Spirilla: *Spirillum volutans*, *Spirillum minus*
- Spirochetes: *Treponema pallidum*, *Borrelia burgdorferi*

## Structure and Function of Cells

- Cells are the basic units of life that contain the essential molecules and organelles for performing various functions.
- Cells can be classified into two major types: prokaryotic and eukaryotic.
- Prokaryotic cells are simpler and smaller than eukaryotic cells. They lack a nucleus and other membrane-bound organelles. They are mostly unicellular organisms, such as bacteria and archaea.
- Eukaryotic cells are more complex and larger than prokaryotic cells. They have a nucleus and other membrane-bound organelles, such as mitochondria, chloroplasts, endoplasmic reticulum, Golgi apparatus, lysosomes, etc. They are mostly multicellular organisms, such as plants, animals, fungi, and protists.

## Size, Shape, and Arrangement of Bacterial Cells

- Bacterial cells are prokaryotic cells that vary in size, shape, and arrangement.
- The size of bacterial cells ranges from 0.2 to 10 micrometers in diameter.
- The shape of bacterial cells can be classified into three main types: cocci (spherical), bacilli (rod-shaped), and spirilla (spiral-shaped).
- The arrangement of bacterial cells can be influenced by their shape and mode of division. Some common arrangements are: diplococci (pairs of cocci), streptococci (chains of cocci), staphylococci (clusters of cocci), diplobacilli (pairs of bacilli), streptobacilli (chains of bacilli), and vibrios (comma-shaped bacilli).

## Structure and Function of Cell Membrane

- The cell membrane is a thin layer of phospholipids and proteins that surrounds the cytoplasm of a cell.
- The cell membrane acts as a selective barrier that regulates the entry and exit of substances into and out of the cell.
- The cell membrane also maintains the shape and integrity of the cell, and facilitates cell communication and recognition.

## Structure and Function of Cytoplasm

- The cytoplasm is a jelly-like substance that fills the interior of a cell.
- The cytoplasm contains various organelles, molecules, and ions that are involved in cellular functions.
- The cytoplasm also allows the movement of substances and organelles within the cell, which is called cyclosis or cytoplasmic streaming.

## **Structure and Function of Nucleus**

- The nucleus is a membrane-bound organelle that is present only in eukaryotic cells.
- The nucleus contains the genetic material (DNA) of the cell, which is organized into chromosomes.
- The nucleus controls the expression of genes and the synthesis of proteins, which are essential for cellular functions.
- The nucleus also contains a smaller structure called the nucleolus, which is responsible for the production of ribosomes.

## **Structure and Function of Ribosomes**

- Ribosomes are small structures that are composed of RNA and proteins.
- Ribosomes can be found either free in the cytoplasm or attached to the endoplasmic reticulum.
- Ribosomes are the sites of protein synthesis, where they translate the genetic information from mRNA into amino acid sequences.

## **Structure and Function of Mitochondria**

- Mitochondria are membrane-bound organelles that are present in most eukaryotic cells.
- Mitochondria have a double membrane structure, with an inner membrane that is folded into cristae and an outer membrane that is smooth.
- Mitochondria are the sites of cellular respiration, where they convert glucose and oxygen into ATP (energy) and carbon dioxide.
- Mitochondria also have their own DNA and ribosomes, and can replicate independently of the nucleus.

## **Structure and Function of Chloroplasts**

- Chloroplasts are membrane-bound organelles that are present only in plant cells and some algae.
- Chloroplasts have a double membrane structure, with an inner membrane that contains flattened sacs called thylakoids and an outer membrane that is smooth.
- Chloroplasts are the sites of photosynthesis, where they convert light energy and carbon dioxide into glucose and oxygen.
- Chloroplasts also have their own

### **Unit 3 - Catabolic & anabolic reactions: enzymes, energy production and carbohydrate metabolism. Lipid & protein catabolism, Energy production mechanism, metabolic diversity & pathways of energy use. Integration of metabolism**

- Catabolic and anabolic reactions are two types of metabolic reactions that occur in living cells.
- Catabolic reactions break down complex molecules into simpler ones, releasing energy and reducing power (e.g., ATP, NADH, FADH<sub>2</sub>) that can be used for other cellular processes .
- Examples of catabolic reactions are glycolysis, the citric acid cycle, oxidative phosphorylation, fatty acid oxidation, and amino acid degradation .
- Anabolic reactions use energy and reducing power to synthesize complex molecules from simpler ones, such as proteins, carbohydrates, lipids, and nucleic acids .
- Examples of anabolic reactions are gluconeogenesis, glycogenesis, the pentose phosphate pathway, fatty acid synthesis, and amino acid synthesis .
- Enzymes are biological catalysts that speed up the rate of metabolic reactions by lowering the activation energy required for the reactions to occur .
- Enzymes are usually proteins that have specific active sites where substrates bind and undergo chemical transformation .
- Enzymes can be regulated by various factors, such as temperature, pH, substrate concentration, inhibitors, activators, and feedback mechanisms .
- Energy production is the process of converting the chemical energy stored in fuel molecules (such as glucose, fatty acids, and amino acids) into a more usable form of energy (such as ATP) that can power cellular work .
- Energy production involves three main stages: glycolysis, the citric acid cycle, and oxidative phosphorylation .
- Glycolysis is the breakdown of glucose into two molecules of pyruvate, generating two molecules of ATP and two molecules of NADH per glucose molecule .
- The citric acid cycle is the oxidation of acetyl-CoA (derived from pyruvate or fatty acids) into carbon dioxide, generating two molecules of ATP, six molecules of NADH, and two molecules of FADH<sub>2</sub> per acetyl-CoA molecule .
- Oxidative phosphorylation is the transfer of electrons from NADH and FADH<sub>2</sub> to oxygen, generating a proton gradient across the inner mitochondrial membrane that drives the synthesis of ATP by ATP synthase .

- Carbohydrate metabolism is the set of metabolic pathways that involve the synthesis and degradation of carbohydrates, such as glucose, glycogen, and pentoses .
- Carbohydrate metabolism includes catabolic pathways, such as glycolysis, the citric acid cycle, and the pentose phosphate pathway, and anabolic pathways, such as gluconeogenesis and glycogenesis .
- Lipid catabolism is the breakdown of lipids, such as triglycerides and fatty acids, into acetyl-CoA, glycerol, and ketone bodies, which can be used as energy sources or substrates for other metabolic pathways [

## Catabolic and Anabolic Reactions

- Catabolic and anabolic reactions are two types of metabolic reactions that occur in living cells.
- Catabolic reactions are those that break down complex molecules into simpler ones, releasing energy in the process. Examples of catabolic reactions are glycolysis, beta-oxidation, and proteolysis.
- Anabolic reactions are those that build up complex molecules from simpler ones, using energy in the process. Examples of anabolic reactions are protein synthesis, photosynthesis, and lipogenesis.
- Catabolic and anabolic reactions are often coupled, meaning that the energy released by catabolic reactions is used to power anabolic reactions. A common molecule that mediates this coupling is adenosine triphosphate (ATP), which is synthesized by catabolic reactions and hydrolyzed by anabolic reactions.
- Catabolic and anabolic reactions are regulated by various factors, such as hormones, enzymes, and feedback mechanisms, to maintain the balance between energy production and consumption, and between the synthesis and degradation of biomolecules.

## Enzymes

- Enzymes are biological catalysts that speed up the rate of biochemical reactions by lowering the activation energy required for the reaction to occur.
- Enzymes are highly specific for their substrates, which are the molecules that bind to the enzyme's active site and undergo a chemical change.
- Enzymes can be classified into two main types based on the direction of the reaction they catalyze: anabolic enzymes and catabolic enzymes.

## Anabolic Enzymes

- Anabolic enzymes are involved in the synthesis of larger and more complex molecules from smaller units.

- Anabolic reactions are also called biosynthetic reactions, as they produce new biomolecules that are essential for the growth and maintenance of cells.
- Anabolic reactions consume energy, as they require input of ATP or other high-energy molecules to drive the formation of bonds.
- An example of an anabolic reaction is photosynthesis, where plants use light energy to convert carbon dioxide and water into glucose and oxygen.

## Catabolic Enzymes

- Catabolic enzymes are involved in the breakdown of larger and more complex molecules into smaller units.
- Catabolic reactions are also called degradative reactions, as they degrade biomolecules that are no longer needed or that can be used as energy sources.
- Catabolic reactions release energy, as they break bonds and generate ATP or other high-energy molecules that can be used for cellular work.
- An example of a catabolic reaction is cellular respiration, where cells use oxygen to oxidize glucose and produce carbon dioxide, water and ATP.

## Metabolic Pathways

- Metabolic pathways are sequences of enzyme-catalyzed reactions that transform one or more substrates into one or more products.
- Metabolic pathways can be either linear, where the product of one reaction is the substrate for the next, or cyclic, where the product of the last reaction is the substrate for the first.
- Metabolic pathways can be either reversible, where the reaction can proceed in both directions, or irreversible, where the reaction can only proceed in one direction.
- Metabolic pathways can be either anabolic, catabolic, or amphibolic, where they involve both anabolic and catabolic reactions.
- Metabolic pathways are regulated by various factors, such as the availability of substrates and products, the activity and expression of enzymes, and the feedback inhibition or activation of enzymes by the end products or intermediates of the pathway.

## **Energy production for the notes of the Unit 3 - Catabolic & anabolic reactions: enzymes, energy production and carbohydrate metabolism. Lipid & protein catabolism, Energy production mechanism, metabolic diversity & pathways of energy use. Integration of metabolism in the subject of REMEDIAL BIOLOGY-II KCS**

### **Catabolic and anabolic reactions**

- Catabolic reactions are those that break down large molecules into smaller ones and release energy. An example of a catabolic reaction is the process of food digestion, where different enzymes break down food particles so they can be absorbed by the small intestine.
- Anabolic reactions are those that build up large molecules from smaller ones and require energy. An example of an anabolic reaction is photosynthesis, where plants make glucose molecules from different raw materials.
- Catabolic and anabolic reactions are connected by the common currency of energy, which is ATP. ATP is produced by catabolic reactions and used by anabolic reactions.
- Catabolic and anabolic reactions are also regulated by enzymes, which are proteins that catalyze or speed up chemical reactions. Enzymes lower the activation energy of a reaction, which is the amount of energy needed to start the reaction.
- Enzymes have specific shapes and active sites that bind to specific substrates or reactants. The enzyme-substrate complex undergoes a chemical change and produces products that are released from the enzyme. The enzyme can then bind to another substrate and repeat the process.
- Enzymes can be affected by factors such as temperature, pH, concentration, inhibitors, and activators. These factors can change the shape or activity of the enzyme and affect the rate of the reaction.

### **Energy production and carbohydrate metabolism**

- Carbohydrates are organic molecules that consist of carbon, hydrogen, and oxygen atoms. They are the main source of energy for living organisms and can be classified into monosaccharides, disaccharides, and polysaccharides based on the number of sugar units they contain.
- Monosaccharides are the simplest carbohydrates and include glucose, fructose, and galactose. They can be joined together by glycosidic bonds to form disaccharides, such as sucrose, lactose, and maltose. Disaccharides can be further joined to form polysaccharides, such as starch, glycogen, cellulose, and chitin.

- Carbohydrate metabolism is the process of breaking down or synthesizing carbohydrates to produce or store energy. The most common pathway of carbohydrate metabolism is glycolysis, which is the catabolic breakdown of glucose into two molecules of pyruvate. Glycolysis occurs in the cytoplasm of the cell and produces a net gain of two ATP and two NADH molecules per glucose molecule.
- Pyruvate can then enter different pathways depending on the availability of oxygen and the type of organism. In the presence of oxygen, pyruvate is transported into the mitochondria and converted into acetyl-CoA, which enters the citric acid cycle (also known as the Krebs cycle or the tricarboxylic acid cycle). The citric acid cycle is a series of reactions that produces two ATP, six NADH, and two FADH<sub>2</sub> molecules per acetyl-CoA molecule. The NADH and FADH<sub>2</sub> molecules then transfer their electrons to the electron transport chain, which is a series of protein complexes embedded in the inner mitochondrial membrane. The electron transport chain generates a proton gradient across the membrane, which drives the synthesis of ATP by the enzyme ATP synthase. This process is called oxidative phosphorylation and produces about 28 ATP molecules per glucose molecule.
- In the absence of oxygen, pyruvate undergoes fermentation, which is the anaerobic conversion of pyruvate into other molecules that can regenerate NAD<sup>+</sup> for glycolysis. There are different types of fermentation, such as lactic acid fermentation and alcoholic fermentation. Lactic acid fermentation occurs in some animal cells and some bacteria, and converts pyruvate into lactate. Alcoholic fermentation occurs in some plants and fungi, and converts pyruvate into ethanol and carbon dioxide.
- Carbohydrate metabolism can also involve the synthesis of carbohydrates from other molecules, such as amino acids, glycerol, and lactate. This process is called gluconeogenesis and occurs mainly in the liver. Gluconeogenesis is the reverse of glycolysis,

## Carbohydrate Metabolism

- Carbohydrate metabolism is the whole of the biochemical processes responsible for the metabolic formation, breakdown, and interconversion of carbohydrates in living organisms.
- Carbohydrates are organic molecules composed of carbon, hydrogen, and oxygen atoms. They are the main source of energy for most living cells and can be classified into simple sugars (monosaccharides) and complex sugars (polysaccharides).
- Carbohydrate metabolism begins with digestion in the small intestine where monosaccharides are absorbed into the blood stream. Blood sugar concentrations are controlled by three hormones: insulin, glucagon, and epinephrine.
- The major pathways of carbohydrate metabolism are:

- Glycolysis: the breakdown of glucose into pyruvate, which can be further oxidized in the citric acid cycle or converted into lactate or ethanol under anaerobic conditions. Glycolysis produces a net gain of 2 ATP and 2 NADH molecules per glucose molecule.
- Citric acid cycle: the oxidation of acetyl-CoA (derived from pyruvate or fatty acids) into carbon dioxide and water, generating 3 NADH, 1 FADH<sub>2</sub>, and 1 GTP per cycle. The citric acid cycle is the central hub of cellular metabolism and connects with other pathways such as amino acid and lipid metabolism.
- Oxidative phosphorylation: the transfer of electrons from NADH and FADH<sub>2</sub> to oxygen, mediated by a series of protein complexes in the inner mitochondrial membrane. The electron transport chain creates a proton gradient that drives the synthesis of ATP by ATP synthase. Oxidative phosphorylation produces about 28 ATP molecules per glucose molecule.
- Pentose phosphate pathway: the generation of pentose sugars (such as ribose) and NADPH from glucose. The pentose phosphate pathway is important for nucleotide synthesis and antioxidant defense.
- Glycogenesis: the synthesis of glycogen from glucose. Glycogen is a branched polymer of glucose that serves as a storage form of glucose in the liver and muscle cells.
- Glycogenolysis: the breakdown of glycogen into glucose. Glycogenolysis is stimulated by glucagon and epinephrine when blood glucose levels are low.
- Gluconeogenesis: the synthesis of glucose from non-carbohydrate precursors, such as lactate, pyruvate, glycerol, and amino acids. Gluconeogenesis occurs mainly in the liver and is essential for maintaining blood glucose levels during fasting or starvation.
- Carbohydrate metabolism is regulated by various factors, such as hormonal signals, enzyme activity, substrate availability, and feedback inhibition. Carbohydrate metabolism is also influenced by the metabolic diversity and pathways of energy use of different cell types and tissues.
- Carbohydrate metabolism is integrated with lipid and protein metabolism, as they share common intermediates and cofactors. For example, glycerol from triglycerides can be converted into glucose by gluconeogenesis, and amino acids from proteins can be converted into acetyl-CoA by deamination and decarboxylation.

## Lipid and Protein Catabolism

- Lipid and protein catabolism are the processes of breaking down lipids and proteins, respectively, into smaller molecules that can be used for energy production or biosynthesis.
- Lipids and proteins are important macromolecules that store energy, provide structural support, and perform various functions in living organisms.

- Lipid and protein catabolism involve different enzymes, pathways, and intermediates, but they are connected to the central catabolic pathways of glucose metabolism, such as glycolysis, the citric acid cycle, and oxidative phosphorylation.

## Lipid Catabolism

- Lipids are a diverse group of organic molecules that include fats, oils, phospholipids, steroids, and waxes.
- The most common form of lipid catabolism is the breakdown of triglycerides, which are composed of glycerol and three fatty acids.
- Triglycerides are hydrolyzed by lipases, enzymes that cleave the ester bonds between glycerol and fatty acids .
- Glycerol can be converted to dihydroxyacetone phosphate (DHAP), an intermediate of glycolysis, by a series of reactions involving glycerol kinase and glycerol-3-phosphate dehydrogenase .
- Fatty acids are transported to the mitochondria by a carrier protein called carnitine, where they undergo beta-oxidation, a cyclic process that removes two-carbon units from the fatty acid chain and produces acetyl-CoA, NADH, and FADH<sub>2</sub> .
- Acetyl-CoA can enter the citric acid cycle and generate more NADH, FADH<sub>2</sub>, and ATP, while NADH and FADH<sub>2</sub> can donate electrons to the electron transport chain and drive oxidative phosphorylation .
- Lipid catabolism can produce more energy per gram than carbohydrate catabolism, because lipids have more carbon-hydrogen bonds and less oxygen atoms than carbohydrates.

## Protein Catabolism

- Proteins are polymers of amino acids, which are linked by peptide bonds.
- Protein catabolism begins with the hydrolysis of proteins into amino acids by proteases, enzymes that cleave the peptide bonds .
- Amino acids can be transported into the cells by specific transporters, where they undergo deamination, the removal of the amino group, which is converted to ammonia and then to urea for excretion .
- The remaining carbon skeleton of the amino acid can be converted to various intermediates of glucose metabolism, such as pyruvate, acetyl-CoA, or citric acid cycle intermediates, depending on the type of amino acid .
- These intermediates can then be used for energy production or biosynthesis of other molecules, such as glucose, fatty acids, or ketone bodies .
- Protein catabolism is usually a secondary source of energy, as proteins are mainly used for structural and functional purposes in living organisms. However, protein catabolism can increase under conditions of starvation, stress, or disease, when the body needs to mobilize its energy reserves.

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## Energy production mechanism

- Energy production is the process of converting energy from one form to another in living systems.
- Energy comes to living systems through electrons occupying high energy states, either from food (respiratory chains) or from light (photosynthesis).
- Energy is exchanged between living systems and their surroundings as they use energy from the sun to perform photosynthesis or consume energy-storing molecules and release energy to the environment by doing work and releasing heat.
- Energy is subject to physical laws, such as the first law of thermodynamics, which states that energy cannot be created or destroyed, only transformed, and the second law of thermodynamics, which states that the entropy (disorder) of the universe tends to increase.
- The primary energy-releasing pathways of metabolism involve the breakdown and synthesis of carbohydrates, lipids, and amino acids.
- Carbohydrates are the most important energy source for most living organisms, as they can be easily broken down into glucose, which is then oxidized to produce ATP (adenosine triphosphate), the main energy currency of the cell.
- Lipids and proteins can also be used as energy sources, but they require more steps to be converted into glucose or other intermediates that can enter the metabolic pathways.
- The main metabolic pathway for energy production from glucose is glycolysis, which occurs in the cytoplasm of the cell and produces two molecules of pyruvate, two molecules of ATP, and two molecules of NADH (nicotinamide adenine dinucleotide) per molecule of glucose.
- Pyruvate can then enter the mitochondria, where it is converted into acetyl-CoA, which enters the citric acid cycle (also known as the Krebs cycle or the tricarboxylic acid cycle), which produces more ATP, NADH, and FADH<sub>2</sub> (flavin adenine dinucleotide).
- The electrons from NADH and FADH<sub>2</sub> are then transferred to the electron transport chain, which is located in the inner membrane of the mitochondria, where they power the synthesis of ATP by oxidative phosphorylation.
- The final electron acceptor in the electron transport chain is oxygen, which

is reduced to water.

- The total amount of ATP produced from one molecule of glucose by aerobic respiration (involving oxygen) is about 36 to 38.
- In some organisms, such as bacteria, archaea, and some eukaryotes, anaerobic respiration (not involving oxygen) can also occur, using different electron acceptors, such as nitrate, sulfate, or carbon dioxide.
- In some conditions, such as low oxygen availability, pyruvate can be converted into lactate or ethanol by fermentation, which regenerates NAD<sup>+</sup> from NADH, allowing glycolysis to continue, but producing less ATP than aerobic respiration.
- Energy production from light is carried out by photosynthesis, which occurs in the chloroplasts of plants, algae, and some bacteria.
- Photosynthesis consists of two stages: the light reactions and the Calvin cycle.
- The light reactions use light energy to split water molecules and produce ATP and NADPH, which are then used to power the Calvin cycle.
- The Calvin cycle uses carbon dioxide and the energy from ATP and NADPH to synthesize glucose and other organic molecules, which can then be used for energy storage or biosynthesis.
- The energy stored in organic molecules can be passed to other organisms in the ecosystem when those organisms eat plants (or eat other organisms that have previously eaten plants).
- Metabolic diversity refers to the variety of ways that different organisms can obtain and use energy from different sources and in different environments.
- Metabolic pathways of energy use are the specific sequences of biochemical reactions that convert energy from one form to another in living systems.
- Integration of metabolism is the coordination and regulation of metabolic pathways to maintain homeostasis and respond to environmental changes.
- Metabolic pathways are regulated by various factors, such as enzyme activity, substrate availability, feedback inhibition, allosteric modulation, hormonal signals, and gene expression.
- Metabolic pathways are interconnected and interdependent, as the products of one pathway can be the substrates of another, or the intermediates of multiple pathways.
- Metabolic pathways can be classified

## Metabolic Diversity and Pathways of Energy Use

- Metabolic diversity mainly refers to the different metabolic strategies that organisms have evolved to obtain energy.
- Metabolic pathways are series of connected chemical reactions that feed one another and convert starting molecules into products.
- Metabolic pathways can be broadly divided into two categories based on their effects:

- Catabolic pathways break down complex molecules into simpler ones and release energy. Examples are glycolysis, Krebs cycle, and oxidative phosphorylation.
  - Anabolic pathways synthesize complex molecules from simpler ones and require energy. Examples are photosynthesis, gluconeogenesis, and protein synthesis.
- Metabolic pathways are also classified based on the source of energy and carbon for the organisms:
  - Phototrophs use light as the energy source and can be further divided into photoautotrophs (use CO<sub>2</sub> as the carbon source) and photoheterotrophs (use organic compounds as the carbon source).
  - Chemotrophs use chemical compounds as the energy source and can be further divided into chemoautotrophs (use CO<sub>2</sub> as the carbon source) and chemoheterotrophs (use organic compounds as the carbon source).
- Metabolic pathways are regulated by various factors, such as enzyme activity, substrate availability, feedback inhibition, and hormonal signals.
- Metabolic pathways are interconnected and integrated to maintain homeostasis and respond to changing environmental conditions.
- Metabolic pathways evolved among prokaryotes before eukaryotes arose as the result of their interaction and coevolution with changing physico-chemical environmental conditions.

## Integration of metabolism

- Metabolism is the sum of all the chemical reactions that take place in a living cell to maintain its structure and function.
- Metabolism can be divided into two major types: catabolism and anabolism.
  - Catabolism is the breakdown of complex molecules into simpler ones, releasing energy in the process.
  - Anabolism is the synthesis of complex molecules from simpler ones, requiring energy in the process.
- Metabolism is regulated by various factors, such as enzymes, hormones, nutrients, and environmental conditions .
  - Enzymes are biological catalysts that speed up the rate of metabolic reactions by lowering the activation energy.
  - Hormones are chemical messengers that coordinate the metabolic activities of different tissues and organs.
  - Nutrients are the sources of energy and building blocks for metabolic reactions.
  - Environmental conditions, such as temperature, pH, and oxygen availability, affect the activity and stability of enzymes and other metabolic components.
- Metabolism is integrated by the interconnection and coordination of dif-

ferent metabolic pathways .

- A metabolic pathway is a series of connected chemical reactions that feed one another, converting one or more starting molecules into products.
- Metabolic pathways can be broadly divided into two categories based on their effects:
  - \* Catabolic pathways are those that break down molecules and release energy, such as glycolysis, lipid catabolism, and protein catabolism.
  - \* Anabolic pathways are those that build up molecules and require energy, such as gluconeogenesis, lipid synthesis, and protein synthesis.
- Metabolic pathways are interconnected by common intermediates, such as acetyl-CoA, pyruvate, and oxaloacetate, that can be used for different purposes depending on the metabolic needs of the cell .
- Metabolic pathways are coordinated by feedback mechanisms, such as allosteric regulation, covalent modification, and hormonal control, that ensure the balance between energy production and consumption, and between biosynthesis and degradation .
- Metabolism is diverse and adaptable, as different organisms and tissues can use different sources and pathways of energy depending on their environment and function .
  - Energy production can occur through different mechanisms, such as substrate-level phosphorylation, oxidative phosphorylation, and photophosphorylation, depending on the availability of oxygen and light .
  - Energy use can occur through different pathways, such as the pentose phosphate pathway, the citric acid cycle, and the urea cycle, depending on the need for reducing power, biosynthetic precursors, and nitrogen disposal .

## Unit 4 - Energy Utilization

- Structure of mitochondria
  - Mitochondria are small, rod-shaped organelles found in most eukaryotic cells, such as plants and animals .
  - They have two membranes, an outer membrane and an inner membrane, made of phospholipids .
  - The inner membrane is folded into projections called cristae, which increase the surface area for chemical reactions .
  - The space between the two membranes is called the intermembrane space, and the space inside the inner membrane is called the matrix .
  - The matrix contains enzymes, DNA, ribosomes, and various molecules involved in energy production .
  - Mitochondria are the sites of cellular respiration, a process that con-

- verts organic molecules into ATP, the main energy currency of the cell .
- Mitochondria have their own DNA and can replicate independently of the cell .
- Mitochondria are thought to have evolved from ancient bacteria that were engulfed by ancestral eukaryotic cells in a symbiotic relationship .
- Cellular respiration
  - Cellular respiration is a series of metabolic reactions that break down glucose and other organic molecules into carbon dioxide and water, releasing energy in the form of ATP .
  - Cellular respiration can be divided into four stages: glycolysis, formation of acetyl co-A, Krebs cycle, and electron transport chain .
  - Glycolysis is the first stage of cellular respiration, which occurs in the cytoplasm of the cell. It splits one molecule of glucose into two molecules of pyruvate, producing two molecules of ATP and two molecules of NADH (a coenzyme that carries electrons) per glucose molecule .
  - Formation of acetyl co-A is the second stage of cellular respiration, which occurs in the mitochondria. It converts each pyruvate molecule into a two-carbon molecule called acetyl co-A, releasing one molecule of carbon dioxide and one molecule of NADH per pyruvate molecule .
  - Krebs cycle is the third stage of cellular respiration, which occurs in the matrix of the mitochondria. It completes the oxidation of glucose by breaking down each acetyl co-A molecule into two molecules of carbon dioxide, producing one molecule of ATP, three molecules of NADH, and one molecule of FADH<sub>2</sub> (another coenzyme that carries electrons) per acetyl co-A molecule .
  - Electron transport chain is the fourth and final stage of cellular respiration, which occurs in the inner membrane of the mitochondria. It uses the electrons carried by NADH and FADH<sub>2</sub> to pump protons across the inner membrane, creating a gradient of protons in the intermembrane space. The protons then flow back into the matrix through a protein complex called ATP synthase, which uses the energy to synthesize ATP from ADP and phosphate. This process is also called oxidative phosphorylation, because it involves the transfer of electrons to oxygen, the final electron acceptor, forming water .
  - Cellular respiration can produce up to 38 molecules of ATP per glucose molecule, depending on the efficiency of the process and the availability of oxygen .
- Relationship of carbohydrate metabolism to other compounds
  - Carbohydrate metabolism is the process of converting carbohydrates, such as glucose, into energy or other molecules, such as amino acids, fatty acids, and nucleotides .
  - Carbohydrate metabolism is interconnected with other metabolic

pathways, such as protein metabolism and lipid metabolism, through various intermediates and products .

- For example, some amino acids can be synthesized from glycolysis intermediates, such as pyruvate and 3-phosphoglycerate, or from Krebs cycle intermediates, such as oxaloacetate and alpha-ketoglutarate .
- Similarly, some fatty acids can be synthesized from acetyl co-A, the product of

## Energy Utilization

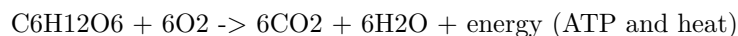
- Energy utilization is the process of converting chemical energy into useful work by living organisms.
- Energy utilization involves the breakdown of organic molecules, such as glucose, by cellular respiration, which releases energy that can be used to synthesize ATP, the primary energy currency of all cells.
- Energy utilization also involves the relationship of carbohydrate metabolism to other compounds, such as lipids, proteins, and nucleic acids, which can be used as alternative sources of energy or as building blocks for biosynthesis.

## Structure of mitochondria

- Mitochondria are organelles that are the site of most of the energy production in eukaryotic cells.
- Mitochondria have a double membrane structure: an outer membrane that surrounds the organelle and an inner membrane that forms folds called cristae, which increase the surface area for energy production.
- Mitochondria also have a matrix, which is the space enclosed by the inner membrane, where some of the enzymes involved in cellular respiration are located.
- Mitochondria have their own DNA and ribosomes, which allow them to produce some of their own proteins.

## Cellular respiration

- Cellular respiration is the process of oxidizing organic molecules, such as glucose, to carbon dioxide and water, and releasing energy that can be used to synthesize ATP.
- Cellular respiration consists of three main stages: glycolysis, the citric acid cycle (also known as the Krebs cycle), and the electron transport chain and oxidative phosphorylation.
- Cellular respiration can be summarized by the following equation:

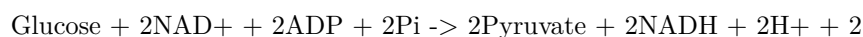


## Relationship of carbohydrate metabolism to other compounds

- Carbohydrate metabolism is the breakdown and synthesis of glucose and other sugars, which are the main source of energy for most cells.
- Carbohydrate metabolism is linked to other compounds, such as lipids, proteins, and nucleic acids, by various metabolic pathways that can interconvert different molecules depending on the cellular needs and availability of substrates.
- For example, excess glucose can be stored as glycogen in the liver and muscles, or converted to fatty acids and stored as triglycerides in adipose tissue. Conversely, when glucose is scarce, glycogen can be broken down to glucose, or fatty acids can be oxidized to acetyl-CoA, which can enter the citric acid cycle.
- Similarly, amino acids, the building blocks of proteins, can be used as a source of energy by deamination, which removes the amino group and produces ammonia, which is excreted as urea. The remaining carbon skeleton can be converted to pyruvate, acetyl-CoA, or other intermediates of the citric acid cycle, depending on the type of amino acid. Conversely, when amino acids are needed for protein synthesis, they can be synthesized from glucose or other intermediates by transamination, which transfers the amino group from one molecule to another.
- Nucleic acids, such as DNA and RNA, can also be degraded and recycled by nucleases, which break down the phosphodiester bonds between the nucleotides, and nucleosidases, which cleave the glycosidic bonds between the nitrogenous bases and the pentose sugars. The resulting nucleosides and nucleobases can be reused for nucleic acid synthesis, or further degraded to produce ribose-5-phosphate, which can enter the pentose phosphate pathway, or uric acid, which is excreted as a waste product.

## Glycolysis

- Glycolysis is the first stage of cellular respiration, which occurs in the cytoplasm of the cell.
- Glycolysis is the process of splitting one molecule of glucose (a six-carbon sugar) into two molecules of pyruvate (a three-carbon compound).
- Glycolysis consists of ten enzymatic reactions, which can be divided into two phases: the preparatory phase and the payoff phase.
- The preparatory phase consumes two molecules of ATP to phosphorylate glucose and its derivatives, making them more reactive and unstable.
- The payoff phase produces four molecules of ATP (a net gain of two ATP) and two molecules of NADH (a reduced form of the electron carrier NAD<sup>+</sup>) by oxidizing the intermediates and transferring their electrons to NAD<sup>+</sup>.
- The overall equation of glycolysis is:



## Structure of mitochondria

- Mitochondria are small, rod-shaped organelles that are found in the cytoplasm of most eukaryotic cells, such as plants and animals .
- They are the sites of cellular respiration, a process that produces ATP, the main energy currency of the cell, from the breakdown of glucose and other organic molecules in the presence of oxygen .
- Mitochondria have a double membrane structure, consisting of an outer membrane and an inner membrane .
  - The outer membrane is smooth and surrounds the entire organelle. It is permeable to small molecules and ions, but not to larger molecules such as proteins .
  - The inner membrane is folded into numerous projections called cristae, which increase the surface area for the reactions of cellular respiration. It is selectively permeable and contains various proteins, such as electron transport chain complexes, ATP synthase, and transporters .
- The space between the two membranes is called the intermembrane space, which contains enzymes and molecules involved in the transfer of electrons and protons during cellular respiration .
- The space enclosed by the inner membrane is called the matrix, which contains the mitochondrial DNA, ribosomes, enzymes, and other molecules involved in the oxidation of pyruvate, the Krebs cycle, and the synthesis of some amino acids and lipids .
- Mitochondria have their own circular DNA, which encodes some of the proteins and RNAs needed for their function. They also have their own ribosomes, which are similar to those of bacteria. This suggests that mitochondria evolved from ancient prokaryotic cells that were engulfed by ancestral eukaryotic cells, a process called endosymbiosis .
- Mitochondria can divide by fission, a process that involves the splitting of the organelle into two daughter mitochondria. They can also fuse with other mitochondria, a process that involves the merging of the membranes and the mixing of the contents. These processes help maintain the number, shape, and function of mitochondria in the cell.

## Cellular Respiration

- Cellular respiration is the process by which biological fuels are oxidised in the presence of an inorganic electron acceptor such as oxygen to produce large amounts of energy, to drive the bulk production of adenosine triphosphate (ATP).
- ATP is the main energy currency of the cell, and it is used to power various cellular processes such as biosynthesis, transport, and movement.
- Cellular respiration can be divided into four main stages: glycolysis, pyruvate oxidation, the citric acid or Krebs cycle, and oxidative phosphorylation.

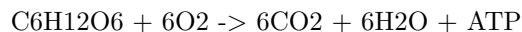
tion.

## Structure of mitochondria

- Mitochondria are the organelles where most of the cellular respiration takes place in eukaryotic cells.
- Mitochondria have a double membrane structure: an outer membrane that surrounds the organelle, and an inner membrane that forms folds called cristae.
- The cristae increase the surface area of the inner membrane, where many of the enzymes and proteins involved in cellular respiration are located.
- The space between the outer and inner membranes is called the intermembrane space, and the space enclosed by the inner membrane is called the matrix.
- The matrix contains the mitochondrial DNA, ribosomes, and some enzymes that catalyse reactions in the citric acid cycle.

## Cellular respiration

- The overall equation for cellular respiration is:

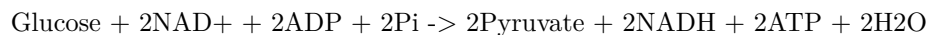


- This means that glucose ( $\text{C}_6\text{H}_{12}\text{O}_6$ ) is oxidised by oxygen ( $\text{O}_2$ ) to produce carbon dioxide ( $\text{CO}_2$ ), water ( $\text{H}_2\text{O}$ ), and ATP.
- The oxidation of glucose involves the transfer of electrons from glucose to oxygen, which is mediated by a series of electron carriers.
- The electron carriers are reduced when they accept electrons, and oxidised when they donate electrons.
- The electron carriers can be divided into two groups: nicotinamide adenine dinucleotide ( $\text{NAD}^+$ ) and flavin adenine dinucleotide (FAD), which are coenzymes that act as mobile electron carriers, and the electron transport chain (ETC), which is a series of protein complexes embedded in the inner mitochondrial membrane that transfer electrons from NADH and  $\text{FADH}_2$  to oxygen.
- The ETC also pumps protons ( $\text{H}^+$ ) from the matrix to the intermembrane space, creating a proton gradient that drives the synthesis of ATP by a protein complex called ATP synthase.

## Glycolysis

- Glycolysis is the first stage of cellular respiration, and it takes place in the cytoplasm of the cell.
- Glycolysis involves the breakdown of one molecule of glucose into two molecules of pyruvate, a three-carbon compound.
- Glycolysis also produces two molecules of ATP and two molecules of NADH per glucose molecule.

- Glycolysis can be divided into two phases: the energy investment phase, where two molecules of ATP are used to phosphorylate glucose and its intermediates, and the energy payoff phase, where four molecules of ATP and two molecules of NADH are generated by the oxidation of the intermediates.
- The net equation for glycolysis is:



- $\text{P}_i$  stands for inorganic phosphate, and  $\text{H}_2\text{O}$  stands for water.

## Pyruvate oxidation

- Pyruvate oxidation is the second stage of cellular respiration, and it takes place in the matrix of the mitochondria.
- Pyruvate oxidation involves the conversion of pyruvate into acetyl coenzyme A (acetyl-CoA), a two-carbon compound that is the main substrate for the citric acid cycle.
- Pyruvate oxidation also produces one molecule of  $\text{CO}_2$  and one molecule of NADH per pyruvate molecule.
- Pyruvate oxidation is catalysed by a multi-enzyme complex called the pyruvate dehydrogenase complex (PDC), which requires several coenzymes and cofactors such as thiamine pyrophosphate (TPP), lipoic acid,

## Relationship of carbohydrate metabolism to other compounds

- Carbohydrates are organic molecules composed of carbon, hydrogen, and oxygen atoms. The family of carbohydrates includes both simple and complex sugars. Glucose and fructose are examples of simple sugars, and starch, glycogen, and cellulose are all examples of complex sugars.
- Carbohydrate metabolism is the process of breaking down carbohydrates into simpler molecules, such as glucose, and using them to generate energy or synthesize other compounds. Carbohydrate metabolism involves several pathways, such as glycolysis, the formation of acetyl CoA, the Krebs cycle, the electron transport system, and oxidative phosphorylation.
- Carbohydrate metabolism is connected to the metabolism of other compounds, such as proteins and lipids, through several intermediates and products. These connections allow the cells to use different substrates for energy production and biosynthesis, depending on the availability and demand.
- The carbohydrates that can also feed into glucose catabolism include galactose, fructose, glycogen, and pentose. These are catabolized during glycolysis, which is the first step of carbohydrate metabolism. Glycolysis

converts glucose into two molecules of pyruvate, which can then enter the mitochondrial matrix and be converted into acetyl CoA.

- The amino acids from proteins can also connect with glucose catabolism through pyruvate, acetyl CoA, and components of the Krebs cycle. The Krebs cycle is the second step of carbohydrate metabolism, which oxidizes acetyl CoA into carbon dioxide and generates reduced coenzymes, such as NADH and FADH<sub>2</sub>. Some amino acids can be deaminated and converted into pyruvate or acetyl CoA, while others can be converted into intermediates of the Krebs cycle, such as oxaloacetate, alpha-ketoglutarate, succinyl CoA, or fumarate .
- The fatty acids from lipids can also connect with glucose catabolism through acetyl CoA. The fatty acids are broken down into acetyl CoA units by a process called beta-oxidation, which occurs in the mitochondrial matrix. The acetyl CoA units can then enter the Krebs cycle and be oxidized further .
- The electron transport system and oxidative phosphorylation are the final steps of carbohydrate metabolism, which use the reduced coenzymes from glycolysis and the Krebs cycle to generate ATP, the main energy currency of the cell. The electron transport system consists of a series of protein complexes and electron carriers that transfer electrons from NADH and FADH<sub>2</sub> to oxygen, forming water. The energy released by this process is used to pump protons across the inner mitochondrial membrane, creating a proton gradient. The oxidative phosphorylation uses this proton gradient to drive the synthesis of ATP by a protein called ATP synthase.
- The ATP generated by carbohydrate metabolism can be used for various cellular processes, such as biosynthesis, transport, movement, and signaling. The ATP can also be stored as glycogen or converted into other compounds, such as nucleotides, amino acids, or lipids.

## Glycolysis

- Glycolysis is a metabolic pathway that converts glucose, a six-carbon sugar, into two molecules of pyruvate, a three-carbon compound.
- Glycolysis occurs in the cytoplasm of the cell and does not require oxygen. It is the first stage of cellular respiration, which produces energy in the form of ATP and NADH.
- Glycolysis consists of ten enzyme-catalyzed reactions that can be divided into two phases: the preparatory phase and the payoff phase.
- In the preparatory phase, two molecules of ATP are used to phosphorylate glucose and split it into two molecules of glyceraldehyde-3-phosphate (G3P).
- In the payoff phase, each G3P molecule is oxidized and phosphorylated, generating two molecules of NADH and four molecules of ATP. The final product is two molecules of pyruvate.
- Glycolysis is an ancient metabolic pathway that is found in most living

organisms. It can provide energy in anaerobic conditions, such as muscle contraction, fermentation, or hypoxia.

- Glycolysis is regulated by several factors, such as the availability of glucose, oxygen, ATP, and NAD<sup>+</sup>, as well as the activity of key enzymes, such as hexokinase, phosphofructokinase, and pyruvate kinase.

## Formation of Acetyl Co-A

- Acetyl Co-A (acetyl coenzyme A) is a molecule that participates in many biochemical reactions in protein, carbohydrate and lipid metabolism.
- Its main function is to deliver the acetyl group to the citric acid cycle (Krebs cycle) to be oxidized for energy production.
- Acetyl Co-A formation occurs inside or outside the cell mitochondria.
- It can be produced via the catabolism (breakdown) of carbohydrates (glucose) and lipids (fatty acids).
- Acetyl Co-A formation most commonly occurs during glucose catabolism.
- After carbohydrates have been broken down by digestive enzymes, the first stage of cellular glucose metabolism or glycolysis can begin.
- Glycolysis is the breaking down of glucose molecules into two molecules of pyruvate, a three-carbon compound, in the cytoplasm.
- Pyruvate then enters the mitochondria, where it is converted into acetyl Co-A by a complex enzyme called pyruvate dehydrogenase.
- The conversion of pyruvate to acetyl Co-A is an irreversible and highly regulated step in cellular respiration.
- The overall formation reaction of acetyl Co-A may be represented as:  
$$\text{pyruvic acid} + \text{CoA} + \text{NAD}^+ \rightarrow \text{acetyl Co-A} + \text{NADH} + \text{H}^+ + \text{CO}_2$$
- This reaction may be called the oxidative decarboxylation of pyruvic acid to acetyl Co-A, because it involves the removal of a carboxyl group (CO<sub>2</sub>) and the transfer of electrons to NAD<sup>+</sup>.
- The acetyl Co-A then combines with a four-carbon molecule called oxaloacetate and goes through a cycle of reactions, ultimately regenerating oxaloacetate.
- This cycle is known as the citric acid cycle or the Krebs cycle, and it takes place in the mitochondrial matrix.
- The citric acid cycle produces ATP (or, in some cases, GTP), NADH, and FADH<sub>2</sub>, which are used for energy production, and releases carbon dioxide as a waste product.

- The NADH and FADH<sub>2</sub> then transfer their electrons to the electron transport chain, which is located in the inner mitochondrial membrane.
- The electron transport chain uses the energy from the electrons to pump protons across the membrane, creating a proton gradient.
- The protons then flow back through a protein called ATP synthase, which uses the energy to synthesize ATP from ADP and phosphate.
- This process is known as oxidative phosphorylation, because it involves the use of oxygen as the final electron acceptor.
- Oxidative phosphorylation is the major source of ATP production in aerobic organisms.
- The formation of acetyl Co-A is a crucial step in cellular respiration, as it links glycolysis to the citric acid cycle and oxidative phosphorylation.
- The formation of acetyl Co-A is also influenced by various factors, such as the availability of oxygen, substrates, and coenzymes, and the activity of enzymes and hormones.
- The formation of acetyl Co-A is regulated by feedback inhibition, allosteric modulation, and covalent modification of the pyruvate dehydrogenase complex.

## Kreb Cycle

The Kreb cycle, also known as the citric acid cycle, is a series of chemical reactions that occur in the mitochondria of aerobic organisms. It is part of the process of cellular respiration, which converts glucose and other organic molecules into energy in the form of ATP .

The main steps of the Kreb cycle are:

- **Step 1:** Pyruvate, the product of glycolysis, is converted into acetyl-CoA by the enzyme pyruvate dehydrogenase complex. This reaction also produces NADH and CO<sub>2</sub> .
- **Step 2:** Acetyl-CoA combines with oxaloacetate, a four-carbon molecule, to form citrate, a six-carbon molecule. This reaction is catalyzed by the enzyme citrate synthase .
- **Step 3:** Citrate is rearranged into isocitrate by the enzyme aconitase. This is a reversible reaction that involves the removal and addition of a water molecule .
- **Step 4:** Isocitrate is oxidized by the enzyme isocitrate dehydrogenase, producing NADH, CO<sub>2</sub>, and alpha-ketoglutarate, a five-carbon molecule. This is the first of the two decarboxylation steps in the cycle .
- **Step 5:** Alpha-ketoglutarate is oxidized by the enzyme alpha-ketoglutarate dehydrogenase, producing NADH, CO<sub>2</sub>, and succinyl-CoA,

a four-carbon molecule. This is the second of the two decarboxylation steps in the cycle .

- **Step 6:** Succinyl-CoA is converted into succinate by the enzyme succinyl-CoA synthetase, releasing a high-energy phosphate group that is used to synthesize ATP or GTP. This is the only step in the cycle that directly produces a nucleoside triphosphate .
- **Step 7:** Succinate is oxidized by the enzyme succinate dehydrogenase, producing FADH<sub>2</sub> and fumarate, a four-carbon molecule. This enzyme is also part of the electron transport chain, where FADH<sub>2</sub> donates its electrons to the complex II .
- **Step 8:** Fumarate is hydrated by the enzyme fumarase, forming malate, a four-carbon molecule. This is another reversible reaction that involves the addition of a water molecule .
- **Step 9:** Malate is oxidized by the enzyme malate dehydrogenase, producing NADH and oxaloacetate, the starting molecule of the cycle. This completes the cycle and allows it to continue with another acetyl-CoA molecule .

The Krebs cycle has several functions and outcomes:

- It generates energy in the form of ATP or GTP, which can be used for various cellular processes .
- It produces electron carriers, such as NADH and FADH<sub>2</sub>, which can donate their electrons to the electron transport chain and drive the synthesis of more ATP by oxidative phosphorylation .
- It releases CO<sub>2</sub> as a waste product of the oxidation of organic molecules .
- It provides intermediates for the synthesis of certain amino acids, such as aspartate, glutamate, and alanine, as well as other compounds, such as heme, purines, and pyrimidines .

The Krebs cycle is influenced by several factors, such as the availability of oxygen, substrates, and enzymes, as well as the demand for ATP and other products. The cycle is regulated by feedback inhibition, allosteric modulation, and covalent modification of some of its key enzymes .

Hello, I am Sydney, your AI assistant. I can help you with your study material. Here are some notes on the topic of Electron Transport System and Oxidative Phosphorylation:

## Electron Transport System and Oxidative Phosphorylation

- Electron Transport System (ETS) is a series of proteins and electron carriers located in the inner mitochondrial membrane that transfers electrons donated by the reduced molecules NADH and FADH<sub>2</sub> through a series of electron acceptors, to oxygen .

- Oxidative Phosphorylation (OP) is a process involving a flow of electrons through the ETS, which creates an electrochemical gradient of protons across the inner mitochondrial membrane, and the use of this gradient to drive the synthesis of ATP by a protein complex called ATP synthase .
- The ETS consists of four major complexes: Complex I (NADH dehydrogenase), Complex II (succinate dehydrogenase), Complex III (cytochrome bc1 complex), and Complex IV (cytochrome c oxidase), as well as two mobile electron carriers: ubiquinone (Q) and cytochrome c (Cyt c) .
- Complex I and II receive electrons from NADH and FADH<sub>2</sub>, respectively, and transfer them to Q, which then delivers them to Complex III. Complex III passes the electrons to Cyt c, which carries them to Complex IV. Complex IV finally transfers the electrons to oxygen, forming water .
- As the electrons move through the ETS, they release energy that is used by Complex I, III, and IV to pump protons from the mitochondrial matrix to the intermembrane space, creating a proton gradient .
- The proton gradient represents a potential energy source that is harnessed by ATP synthase, which allows the protons to flow back to the matrix through its channel. This flow of protons drives the rotation of a part of ATP synthase called the F<sub>1</sub> subunit, which catalyzes the phosphorylation of ADP to ATP .
- The coupling of electron transport and proton pumping to ATP synthesis is called chemiosmosis, and it is the main mechanism of OP .
- OP is the final stage of cellular respiration, and it produces the majority of ATP from the oxidation of glucose and other organic molecules. OP depends on the availability of oxygen as the final electron acceptor, and it can be inhibited by various substances that block the ETS or ATP synthase .

## ATP and Cellular Respiration

- ATP (adenosine triphosphate) is the energy-carrying molecule used in cells because it can release energy very quickly.
- Cellular respiration refers to the breakdown of glucose and other respiratory substrates to make ATP in the presence of an inorganic electron acceptor such as oxygen .
- Cellular respiration is a series of chemical reactions that involve three main steps: glycolysis, the citric acid cycle, and oxidative phosphorylation .
- Glycolysis is the first step of cellular respiration that occurs in the cytoplasm. It splits one molecule of glucose into two molecules of pyruvate, producing a net of 2 ATP and 2 NADH (a reduced form of nicotinamide adenine dinucleotide, an electron carrier) per glucose molecule .
- Pyruvate is then transported into the mitochondria, where it is converted into acetyl coenzyme A (acetyl-CoA) by a complex of enzymes called pyruvate dehydrogenase. This process also produces 1 NADH and 1 CO<sub>2</sub> per pyruvate molecule .

- Acetyl-CoA enters the citric acid cycle (also known as the Krebs cycle or the tricarboxylic acid cycle) in the mitochondrial matrix. It combines with oxaloacetate to form citrate, which then goes through a series of reactions that regenerate oxaloacetate and release 2 CO<sub>2</sub>, 3 NADH, 1 FADH<sub>2</sub> (a reduced form of flavin adenine dinucleotide, another electron carrier), and 1 ATP per acetyl-CoA molecule .
- The NADH and FADH<sub>2</sub> produced in glycolysis, pyruvate oxidation, and the citric acid cycle donate their electrons to the electron transport chain (ETC) in the inner mitochondrial membrane. The ETC consists of a series of protein complexes and electron carriers that transfer the electrons from one to another, releasing energy along the way. The energy is used to pump protons (H<sup>+</sup>) from the mitochondrial matrix to the intermembrane space, creating a proton gradient across the membrane .
- The proton gradient drives the synthesis of ATP by a protein complex called ATP synthase, which allows the protons to flow back into the matrix through its channel. As the protons pass through, they cause the rotation of a part of the ATP synthase that catalyzes the formation of ATP from ADP and a phosphate group. This process is called oxidative phosphorylation because it involves the oxidation of NADH and FADH<sub>2</sub> and the phosphorylation of ADP .
- The final electron acceptor in the ETC is oxygen, which combines with the electrons and protons to form water. This is why cellular respiration requires oxygen and produces water as a by-product .
- The total number of ATP molecules produced by cellular respiration varies depending on the efficiency of the ETC and the transport of NADH and FADH<sub>2</sub> from the cytoplasm to the mitochondria. A common estimate is that one molecule of glucose can produce a net of 30-32 ATP molecules.
- The relationship of carbohydrate metabolism to other compounds is that carbohydrates can be converted into other molecules such as amino acids, fatty acids, and nucleotides, or vice versa, depending on the cellular needs and availability of substrates. These conversions involve various enzymes and intermediates that link the pathways of glycolysis, the citric acid cycle, and oxidative phosphorylation with other metabolic pathways.
- The factors affecting respiration in cells include the availability of oxygen, glucose, and other respiratory substrates, the demand for ATP and other products of respiration, the activity and regulation of the enzymes involved in respiration, and the environmental conditions such as temperature and pH.